

Functional Education

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Narrowing of the subject

Recursive functions without mutable data. (basics of functional programming)

1.1

Learning goals

- Compare two implementations of recursive functions to tell which one follows the idioms of functional programming
- Explain the stop condition and recursion logic of the recursive algorithm
- Design a recursive algorithm for a specific problem based on an iterative algorithm

1.2

1.3

Learning activities sketch

- theory
- compare two or more implementations of recursive functions following functional programming idioms (one correct) and choose the correct one (with AI, explain why one is right and the others are wrong to get personalized feedback to guide the student with hints and explanations)
- blocks to put in the "right order" to make a recursive version of a given iterative algorithm. (with AI, justify the solution and get evaluative feedback, similarly to before, personalized and guided)
- if possible do a final exercise to write a full function in pseudocode without the block system. (maybe python)

1.4

1.5

1.6

1.7

Research question

How does evaluative and personalized feedback of an AI tool support learning of recursive algorithms in functional programming of undergraduate students in scientific topics ?

Target population

Undergraduate students with basics of programming.

1.8

Index of comments

- 1.1 Good choice to narrow the topic down to this.
- 1.2 The first learning goal could use the action verb "Classify" as well.
- 1.3 You should add which specific problem or at least a class of problems to the third learning goal.
- 1.4 Think about how your learning activity support the second learning goal.
- 1.5 Think about which control condition is the fairest to the AI feedback.
- 1.6 Also consider more creative use cases of AI than just as a feedback generator.
- 1.7 You can have a look at Hattie's theory of feedback to be more precise with the terms and think about what kind of feedback is your intervention.
- 1.8 Please be more specific with the target population in terms of age and prior knowledge.